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Document Overview:

- **Structure:** It contains all requested sections including the Abstract, Introduction, Background/Literature Review, Main Thematic and Technical Discussion, Current Approaches, Challenges and Limitations, Research Gaps/Future Directions, Conclusion, and References.
- **Scholarly Support:** It is supported by genuine, recent scientific literature (with persistent identifiers such as DOIs and arXiv IDs), properly cited in-text. It focuses specifically on the limitations of modern large language models in preserving the epistemic integrity (i.e. causal vs. correlational claims) of scientific reporting.
- **Formatting:** The paper is presented as a clean academic manuscript (double-spaced, standard margins) compiled directly through LaTeX to preserve mathematical and scholarly formatting aesthetics.

Preservation of Causal Versus Correlational Language During LLM-Assisted Editing: Risks, Mechanisms, and Epistemic Integrity

Center for Artificial Intelligence in Scientific Communication

August 2026

Abstract

The integration of Large Language Models (LLMs) into the scientific writing process has fundamentally altered how researchers draft, edit, and refine their manuscripts. While LLMs offer significant advantages in fluency, grammar, and stylistic polish, their deployment introduces a critical risk to the epistemic integrity of scientific communication: the inadvertent semantic shift between correlational and causal language. This paper comprehensively examines the preservation of causal versus correlational claims during LLM-assisted editing. We synthesize recent findings demonstrating that LLMs, which operate on probabilistic text generation rather than grounded causal reasoning, frequently distort intended scientific meaning by replacing careful, hedged correlational phrasing with unwarranted causal assertions (or vice versa). By analyzing current literature spanning computational linguistics, epistemology, and biomedical reporting, we identify the technical mechanisms underlying these failures—such as parametric memory override and the lack of robust structural causal models in LLMs. Furthermore, we review current approaches to mitigating these risks, including the development of real-world scientific text datasets and prompt-level constraints. Ultimately, we outline the research challenges, critical limitations of current models, and future directions necessary to ensure that LLM-assisted editing preserves the rigorous causal boundaries essential to scientific discovery.

Keywords: Large Language Models, Scientific Writing, Causal Inference, Epistemic Integrity, Correlational Language, Semantic Shift, AI-Assisted Editing

1 Introduction

The proliferation of Large Language Models (LLMs) has initiated a paradigm shift in academic publishing and scientific communication. Across disciplines, researchers increasingly utilize AI-assisted tools for various stages of the writing process, from initial ideation and drafting to sophisticated developmental editing and proofreading. While the utility of LLMs in enhancing the syntactic fluency and grammatical precision of academic texts is widely acknowledged, their influence on the semantic fidelity—specifically, the precise representation of scientific claims—remains a subject of intense scrutiny (Abdulhai et al., 2026). One of the most critical dimensions of this semantic fidelity is the distinction between causal and correlational language.

In scientific research, particularly within observational studies, biomedicine, and the social sciences, the epistemic boundary between correlation (two variables being associated) and causation (one variable directly influencing another) is foundational. Researchers are trained to deploy specific, often hedged language (e.g., “associated with,” “linked to,” “correlates with”) to accurately reflect the limitations of

observational data, avoiding definitive causal verbs (e.g., “causes,” “determines,” “results in”). However, as LLMs shift from mere post-editing assistants to active co-authors, there is emerging evidence that these models consistently alter the intended meaning of human writing (Abdulhai et al., 2026). When prompted to improve the “flow,” “readability,” or “impact” of a scientific abstract, an LLM might inadvertently upgrade a correlational finding into a causal one, stripping away vital scientific caveats.

This paper investigates the phenomenon of causal language preservation during LLM-assisted editing. We argue that because LLMs lack intrinsic causal reasoning capabilities (Fu et al., 2025) and often fail to accurately infer causal relationships from complex, real-world academic texts (Saklad, 2026), they pose a systemic risk of introducing “spin”—distorted interpretations that mislead readers (Chiu et al., 2017). Through a comprehensive review of the latest developments up to 2026, we explore the theoretical background, technical mechanisms, current mitigation strategies, and future research directions necessary to safeguard the integrity of scientific reporting in the age of AI.

2 Background, Related Work, and Literature Review

2.1 The Epistemology of Causal vs. Correlational Language

The distinction between correlation and causation is not merely a statistical artifact; it is an epistemological cornerstone of the scientific method. Observational studies, which cannot control for all confounding variables, typically yield correlational data. Historically, scientific writing has relied on strict conventions to prevent the overstatement of such findings. However, human readers inherently struggle with this distinction. In a foundational study, Gershman & Ullman (2023) demonstrated that readers frequently draw causal implicatures from purely correlational statements. Even when researchers use orthodox, seemingly non-causal stock phrases (e.g., “X is associated with an increased risk of Y”), laypersons and even domain experts often mentally translate this into “X causes Y.” This innate human bias toward causal narratives makes the precise control of language in scientific manuscripts absolutely paramount; if the text itself introduces causal language where none is warranted, the resultant misinterpretation is magnified.

2.2 ‘Spin’ in Scientific Reporting

The inadvertent or intentional exaggeration of scientific findings is often termed “spin.” Chiu et al. (2017) conducted a methodological systematic review of spin in published biomedical literature, defining it as reporting practices that distort the interpretation of results and mislead readers. They found

that one of the most common manifestations of spin across observational studies and clinical trials is the inappropriate use of causal language to describe mere associations. Spin has severe real-world consequences, potentially misdirecting subsequent clinical research, altering public health policies, and influencing funding allocations based on overstated promises (Chiu et al., 2017). The introduction of LLMs into the drafting process introduces a novel, automated vector for spin, where the exaggeration is not driven by researcher bias, but by algorithmic tendencies toward confident, authoritative-sounding text generation.

2.3 LLMs and the Understanding of Causal Language

Early evaluations of LLMs, such as those conducted on ChatGPT by Kim et al. (2023), revealed significant limitations in the models’ ability to understand causal language in scientific claims. When tasked with labeling the strength of a claim as causal, conditional causal, or correlational, early iterations of ChatGPT lagged considerably behind fine-tuned, domain-specific models. Notably, these models struggled to understand conditional causal claims mitigated by hedges (e.g., “may cause,” “potentially influences”). While subsequent generations of LLMs have grown in parameter size, their fundamental architecture—predicting the next token based on statistical patterns in training data—remains primarily associative rather than causal (Saklad, 2026).

3 Main Thematic and Technical Discussion

3.1 Mechanisms of Semantic Distortion During Editing

When an author prompts an LLM to revise a paragraph containing correlational findings, the model applies learned transformations that optimize for target objectives such as “clarity,” “conciseness,” or “persuasiveness.” Abdulhai et al. (2026) demonstrated that even when LLMs are explicitly prompted to only execute grammar edits based on expert feedback, they still introduce large changes that alter the semantic meaning of the text.

The technical mechanism behind this distortion can be viewed through the lens of what Latif (2026) describes as “Parametric Memory Override” and “Constraint Poisoning.” In high-stakes domain editing, LLMs often ignore explicit user constraints (e.g., “maintain the exact scientific meaning of the associations”) in favor of their parametrically weighted training data. If the model’s training corpus contains a vast amount of text where variables like “smoking” and “lung cancer” are discussed using strong causal verbs, the model will statistically favor those causal verbs during editing, overriding the author’s careful

correlational phrasing. The model conflates statistical likelihood in its training data with scientific truth in the specific context of the manuscript being edited.

3.2 The Absence of Structural Causal Models in LLMs

The failure of LLMs to preserve correlational language is intrinsically linked to their lack of underlying causal reasoning. Fu et al. (2025) systematically analyzed the causal structures of LLMs and Large Reasoning Models (LRMs). By examining Structural Causal Models (SCMs) mapping problem instructions, reasoning steps, and answers, they found that standard LLMs suffer from critical reasoning issues—unfaithfulness and bias—because they rely on superficial correlations. They do not possess a genuine understanding of the physical or logical mechanisms linking two variables. Consequently, when an LLM encounters the phrase “Variable A is linked to Variable B,” it cannot logically assess whether the study design permits a causal claim. It merely acts as a “causal parrot,” generating language that mathematically aligns with the surrounding semantic context, frequently defaulting to more assertive, causal terminology because such language is highly prevalent in declarative texts.

3.3 Evaluation on Real-World Scientific Texts

The inability to preserve causal boundaries is exacerbated by the complexity of real-world academic literature. Saklad (2026) developed a benchmark to test whether LLMs can infer causal relationships from actual scientific texts, which feature varying levels of explicitness, multiple interacting causal events, and complex domains. Their findings indicate that LLMs face significant, perhaps structural, challenges in inferring causal relationships from real-world text, with top models achieving an average F1 score of only 0.535 on such tasks. If an LLM cannot accurately map the causal architecture of a text during the comprehension phase, it is mathematically incapable of reliably preserving that architecture during the generative revision phase.

4 Current Approaches and Developments

4.1 Human-in-the-Loop (HITL) and Constraint Architectures

To mitigate the risks of semantic shift, current developments in LLM-assisted editing emphasize rigid structural controls rather than simple natural-language prompting. Latif (2026) proposes a dual-pass Human-in-the-Loop (HITL) architecture designed to counteract neural network behaviors at the token-generation level. Their research suggests that treating LLMs as autonomous “Auto-Editors” is dangerous

for epistemic integrity. Instead, LLMs must function as “AI Surgical Assistants” operating under the strict, continuous direction of a human domain expert. By utilizing Subject Matter Expert Override Protocols, researchers can lock specific phrases or enforce negative constraints on causal verbs, ensuring the preservation of correlational nuance.

4.2 Dataset Generation for Revisional Modeling

Studying how AI distorts text requires comprehensive baselines of how human scientists actually revise their work. Jourdan et al. (2026) introduced *EarlySciRev*, a dataset of early-stage scientific text revisions extracted from arXiv LaTeX source files. By aligning commented-out discarded text with final published text, researchers have created over 578,000 validated revision pairs grounded in authentic human drafting traces. Datasets like *EarlySciRev* are crucial for the current development pipeline. They allow researchers to evaluate LLM-assisted editing by comparing the model’s suggested revisions against actual human revisions, specifically measuring the rate at which human authors preserve, hedge, or alter causal claims compared to their algorithmic counterparts.

4.3 Advancements in Training Techniques (RLVR)

At the model training level, there are ongoing efforts to instill genuine causal structures into LLM reasoning processes. Fu et al. (2025) observed that models trained with advanced reinforcement learning techniques (such as RLVR) exhibit enhanced causal reasoning capabilities. These training methods help reduce the model’s reliance on spurious correlations and strengthen genuine causal patterns. While these techniques are currently primarily applied to logical reasoning and math tasks, their integration into text-editing models represents a promising avenue for reducing unwarranted causal shifts in generated text.

5 Research Challenges and Limitations

5.1 The Subjectivity of Linguistic Interpretation

A primary challenge in preserving causal versus correlational language is the inherent subjectivity and ambiguity of human language itself. As noted by Kim et al. (2023), choosing appropriate causal language that accurately reflects the strength of evidence is a non-trivial task even for human scientists. Evaluating whether an LLM has inappropriately altered a claim requires robust, objective annotation guidelines, which are notoriously difficult to standardize. The boundary between a “strong correlation”

and a “weak causal link” is often debated within specific scientific sub-domains, making algorithmic alignment incredibly challenging.

5.2 Affective and Psychological Impacts on Authors

The deployment of LLMs as co-authors also introduces psychological limitations that compound the issue of semantic distortion. Zhang (2026) investigated AI-associated academic writing anxiety among postgraduate students, noting that the automation of linguistic execution introduces new evaluative tensions and threats to authorship integrity. When human authors rely heavily on LLMs, they often experience a phenomenon described as “Reactive Writing” (Abdulhai et al., 2026; Zhang, 2026), wherein the author’s primary role shifts from active ideation to passive evaluation of AI suggestions. Under the cognitive load of evaluating rapid AI outputs, authors—particularly novice researchers or non-native speakers—may overlook subtle semantic shifts, inadvertently accepting a causal verb that overstates their correlational data.

5.3 Lack of Established Benchmarks for Tone and Epistemic Shift

While there are robust benchmarks for factual hallucination and grammatical accuracy in LLMs, there is a distinct lack of standardized, automated metrics for evaluating “epistemic shift.” Current NLP evaluation tools struggle to quantify the precise degree to which an edited text has drifted along the correlation-causation continuum. Without large-scale metrics, the scientific community must rely on manual, time-consuming peer review to catch AI-induced spin, a process that is already severely strained.

6 Research Gaps and Future Directions

6.1 Developing Epistemic-Preserving LLMs

Future research must prioritize the development of “Epistemic-Preserving” editing models. This involves fine-tuning LLMs specifically on datasets where edits are strictly constrained by the causal architecture of the original input. Future iterations of models need explicit training on the principles of scientific reporting guidelines (e.g., CONSORT, STROBE) to inherently recognize when variables in an observational study are being discussed and automatically restrict the generative vocabulary to correlational terminology.

6.2 Automated Spin-Detection Agents

There is a critical gap in post-editing verification tools. Future developments should focus on creating specialized, lightweight classifier models—functioning as automated peer reviewers—that analyze differences between original drafts and LLM-edited drafts. These models would highlight any escalation in causal strength, flagging terms like “determines” or “causes” if the underlying methodology identified in the text is purely observational. Integrating such tools into common word processors or LaTeX editors would provide a necessary safeguard against passive semantic shift.

6.3 Longitudinal Studies on Scientific Literature

As AI-assisted writing becomes ubiquitous, longitudinal scientometric studies are required to track shifts in the aggregate causal language of published literature. Future research should mine databases like PubMed and arXiv to determine if the post-2023 era exhibits a statistically significant increase in causal assertions within observational studies. Such empirical evidence will be vital for updating editorial policies at major academic journals.

7 Conclusion

The integration of Large Language Models into scientific writing offers unprecedented efficiency but carries profound risks for the epistemic integrity of scientific communication. This paper has detailed the mechanisms by which LLMs—operating as probabilistic engines devoid of genuine causal reasoning—frequently fail to preserve the crucial distinction between correlational and causal language during the editing process. This semantic distortion acts as an automated form of “spin,” exaggerating scientific claims and potentially misleading readers, policymakers, and subsequent research efforts.

Current models exhibit significant limitations, including parametric memory overrides and an inability to map complex causal relationships in real-world academic texts. While approaches such as Human-in-the-Loop constraints and enhanced reinforcement learning techniques show promise, they require rigorous implementation by vigilant human domain experts. As the scientific community navigates this transition, it is imperative to develop specialized, epistemic-preserving editing models and robust automated spin-detection tools. Ultimately, researchers must recognize that while LLMs can synthesize syntax, the responsibility for scientific causality—and the careful, precise language required to convey it—remains an exclusively human endeavor.

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LAB 1: AI-ASSISTED CITATION INTEGRITY AUDIT

Generate a Research Paper with AI, Then Audit Its Citations

AI Tools for Research • 2 Hours • Individual Activity • Topic IDs T1-T40

Student Detail	Fill in
Name	Yuvraj Singh Rajawat
Roll Number	PHM2026001
Programme / Department	Dual Degree in Human Machine Interaction and Immersive Technologies
Topic ID	T_35
Full Assigned Topic	Preservation of Causal Versus Correlational Language During LLM-Assisted Editing
AI Tool Used	Google Gemini
Date	21/08/2026

IMPORTANT: You will receive a topic from T1-T40. Your task is not to produce the “best” paper. Your task is to generate an AI-written research paper and systematically determine whether its citations can be trusted.

1. PURPOSE OF THIS LAB

Generative AI systems can produce research papers that look scholarly and may contain convincing references, author names, journal titles, DOIs, arXiv IDs, and in-text citations. However, the presence of a citation does not guarantee that the cited publication actually exists, that its metadata are correct, that the identifier belongs to that publication, or that the cited paper supports the statement made by the AI.

In this laboratory activity, you will test these issues yourself by generating a research paper and auditing a systematic sample of its references.

Question 1 - Citation Authenticity: Does the cited publication actually exist, and is it correctly identified?

Question 2 - Claim-Citation Support: If the publication is genuine, does it actually support the claim for which the AI cited it?

2. IMPORTANT RULES

- Use only the topic assigned to you by the instructor.
- Generate the paper in a fresh AI conversation.
- Do not tell the AI that you are conducting a citation audit or hallucination test.
- Do not intentionally ask the AI to generate fake references.
- Preserve the original AI-generated paper exactly as produced.
- Do not correct, regenerate, replace, or question references before saving the original output.
- For verification, do not accept another AI chatbot's statement as proof that a reference exists.
- Use scholarly databases, publisher pages, DOI/arXiv/PMID records, or other reliable scholarly sources.
- Record what you actually find, even if all references are genuine or all are problematic.

Most important principle: You are not being evaluated on whether the AI produces good or bad references. You are being evaluated on how systematically and correctly you audit them.

3. WHAT YOU WILL DO

Part	Activity
A	Record your baseline perception of AI citations
B	Record the AI environment used
C	Generate and preserve an AI-written research paper
D	Create a reference inventory
E	Select 10 references systematically
F	Judge their plausibility before verification
G	Verify authenticity, metadata and identifiers
H	Determine whether genuine references support the AI-generated claims

4. RECOMMENDED 2-HOUR TIME PLAN

Time	Activity
0-10 min	Parts A-B: Baseline and AI environment
10-30 min	Part C: Generate and save research paper
30-40 min	Parts D-E: Inventory and select references
40-45 min	Part F: Pre-verification plausibility
45-80 min	Part G: Authenticity and metadata audit
80-110 min	Part H: Claim-citation entailment audit
110-120 min	Calculate scores, check worksheet and prepare submission

Do not spend excessive time formatting the AI-generated paper. The audit is the main activity.

PART A - BASELINE ASSESSMENT

Complete this section BEFORE opening the AI tool.

A1. How often do you use generative AI for academic work?

☐ Occasionally

A2. Have you previously asked an AI system to generate references?

☐ Never

A3. Have you previously checked whether an AI-generated citation actually exists?

☐ Never

A4. Before this lab, how confident are you that citations generated by AI are generally reliable?

☐ 3 (1 = not at all confident; 5 = completely confident)

Important: Do not change this answer later, even if your opinion changes during the experiment.

PART B - RECORD YOUR AI ENVIRONMENT

Information	Your Response
AI tool used	Google Gemini
Model name	Gemini pro
Model/version shown	3.1
Access tier	☐ Institutional

Web browsing available?	☐ Yes
Was web browsing actually used?	☐ Yes ☐ No ☐ Not sure
Research/Deep Research mode used?	☐ Yes ☐ No
Date/time generated	10 AM 21/08/2026

Important: If the model/version is not displayed, write NOT SHOWN. Do not guess the model version.

PART C - GENERATE THE AI RESEARCH PAPER

C1. Open a Fresh AI Conversation

Do not tell the AI that this is a citation audit, that you are testing hallucinations, or that you expect fabricated references. We want to observe normal AI research-paper generation behaviour.

C2. Use the Following Common Prompt

Replace only [YOUR ASSIGNED TOPIC].

Generate a scholarly research paper on the topic: “[YOUR ASSIGNED TOPIC]”. The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.

C3. Preserve the Original Output

- ☐ Save the complete AI-generated paper
- ☐ Save the complete reference list
- ☐ Take screenshot(s) showing the generated output/references
- ☐ Save conversation/export/share evidence if available

STOP: Do not ask the AI to correct or verify anything yet. Your original output is the experimental evidence.

PART D - PAPER PROFILE AND REFERENCE INVENTORY

Paper title:

Measure	Value
Approximate pages	8
Approximate word count	2300
Number of sections	7
Total number of references	9
Approximate in-text citation occurrences	15
Did AI warn you to verify references?	☐ No

D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

Reference Type	Count
DOI available	5
PMID available, no DOI	
arXiv ID available, no DOI/PMID	1
URL only	
No persistent identifier supplied	3
TOTAL	9

Examples

Category	Example	How to Count
DOI available	Kather, J. N., et al. (2019). Predicting survival from colorectal cancer histology slides using deep learning. PLoS Medicine. DOI: 10.1371/journal.pmed.1002730	DOI available
PMID available, no DOI	A biomedical paper for which the AI provides PMID: 12345678, but no DOI is supplied/found	PMID available, no DOI
arXiv ID available	Yao, S., et al. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629	arXiv ID available
URL only	World Health Organization. Colorectal cancer. Available at: https://... with no DOI, PMID or arXiv ID	URL only
No persistent identifier	Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.	No persistent identifier if none is supplied

Important: Absence of a DOI does not mean a reference is fake. An identifier being present also does not prove authenticity; it must be checked.

PART E - SELECT 10 REFERENCES FOR AUDIT

You are not allowed to choose only suspicious-looking references. This prevents selection bias.

If the bibliography contains 10 or fewer references: audit all references.

If it contains more than 10 references: select the first 3 references, the last 3 references, and 4 references approximately evenly distributed through the middle.

Example: If there are 28 references, a reasonable sample could be 1, 2, 3, 8, 13, 18, 23, 26, 27, 28.

Audit Slot	Original Bibliography Reference No.
R1	1
R2	2
R3	3
R4	4
R5	5
R6	6
R7	7
R8	8
R9	9

Important: From this point onward, R1-R10 refer to your audit sample, not necessarily bibliography references 1-10.

PART F - PLAUSIBILITY TEST

Do this BEFORE searching for any reference.

Look only at the citation as generated by the AI. Ask yourself: “If I had not verified this citation, how convincing would it look to me?”

Score	Meaning
1	Obviously suspicious
2	Somewhat suspicious
3	Unsure
4	Looks genuine
5	Completely convincing

Ref.	Initial Impression	Plausibility 1-5	Predict Genuine?
R1	Very convincing	5	Yes
R2	Very convincing	5	Yes
R3	Very convincing	5	Yes
R4	Very convincing	5	Yes
R5	Looks fake	1	No
R6	Very convincing	5	Yes
R7	Looks fake	1	No
R8	Very convincing	5	Yes
R9	Suspicious	2	No

Example: A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

Critical rule: Never change Part F after completing Part G. Your incorrect predictions are valuable experimental observations.

PART G - CITATION AUTHENTICITY AND METADATA AUDIT

G1. What Are You Checking?

- Does the publication exist?
- Is the title correct?
- Are the authors correct?
- Is the year correct?
- Is the journal/conference/source correct?
- Are volume, issue and pages correct, if supplied?
- Does the DOI/arXiv/PMID belong to this publication?

G2. Verification Procedure

- ☐ Persistent identifier: DOI / arXiv / PMID
- ☐ Publisher or official repository
- ☐ Google Scholar
- ☐ Semantic Scholar / OpenAlex
- ☐ Crossref
- ☐ PubMed or another discipline-specific database
- ☐ Exact-title search

☐ Author + title-keyword search

Not acceptable as sole evidence: “I asked another AI and it said the paper exists.” AI can help you search, but final verification must rely on scholarly or publisher evidence.

G3. Assign One A-E Code

Code	Classification	Meaning
A	VERIFIED	Publication exists and important metadata are correct.
B	WRONG METADATA	Publication exists, but one or more bibliographic details are incorrect.
C	FRANKENSTEIN	Real-looking pieces are combined, but that specific publication does not exist.
D	FABRICATED	No reliable scholarly evidence that the publication exists.
E	IDENTIFIER MISMATCH	DOI/arXiv/PMID is invalid or resolves to a different publication.

G4. Examples

A: AI gives a paper and all details match the publisher record.

B: The paper exists, but the AI gives the wrong publication year or journal.

C: The authors are real, the journal is real, and the topic is plausible, but no such paper with that title/authorship combination exists.

D: You cannot find the paper or a credible scholarly record after systematic searching.

E: The DOI supplied by AI opens an entirely different paper or does not resolve.

G5. Record Your Evidence

Ref.	Publication Found?	Title Match?	Authors Match?	Year Match?	Venue Match?	Identifier Match?
R1	Y	Y	Y	Y	Y	Y
R2	Y	Y	Y	Y	Y	Y
R3	Y	Y	Y	Y	Y	Y
R4	Y	Y	Y	Y	Y	Y
R5	Y	Y	Y	Y	Y	NA
R6	Y	Y	Y	Y	Y	Y
R7	Y	Y	N(only 1 author listed)	Y	Y	NA
R8	Y	Y	N(only 1 author listed)	Y/N	Y	Y
R9	Y	Y	Y	Y	Y	NA

Ref.	Verification Source(s)	Identifier Checked	What You Found
R1	Persistent identifier: DOI / arXiv / PMID	yes	Reference exists and is genuine
R2	Persistent identifier: DOI / arXiv / PMID search	yes	Reference exists and is genuine
R3	Persistent identifier: DOI / arXiv / PMID search	yes	Reference exists and is genuine
R4	Persistent identifier: DOI / arXiv / PMID	yes	Reference exists and is genuine
R5	Exact-title search	NA	Reference exists and is genuine
R6	Persistent identifier: DOI / arXiv / PMID	yes	Reference exists and is genuine
R7	Exact-title search	NA	Reference exists and is genuine
R8	Persistent identifier: DOI / arXiv / PMID	yes	Reference exists and is genuine
R9	Exact-title search	NA	Reference exists and is genuine

G6. Final Classification

Ref.	Final Code	One-Line Reason
R1	A	Reference exists and is genuine
R2	A	Reference exists and is genuine
R3	A	Reference exists and is genuine
R4	A	Reference exists and is genuine
R5	A	Reference exists and is genuine
R6	A	Reference exists and is genuine
R7	A	Reference exists and is genuine
R8	A	Reference exists and is genuine
R9	A	Reference exists and is genuine
R10	A	Reference exists and is genuine

Good reason: B: Article exists, but AI reported 2022 instead of 2020.

Poor reason: B: Details wrong.

PART G - SCORING

Count your final classifications: A = 9 B = 0 C = 0 D = 0 E = 0

Total audited: N = A + B + C + D + E = 9

Code	Points
A	4
B	3
C	1
D	0
E	1

Authenticity Points = 9A + 0B + 0C + 0E

Authenticity Score = [(9A + 0B + 0C + 0E) / (4*9)] x 100

My calculation: 9(4) + 3(0) + 1(0) + 0(0) + 1(0) = 36 points

Authenticity Score = 36 / (4 x 9) x 100 = 100

Worked example: If A=6, B=2, C=1, D=1, E=0, then points = 4(6)+3(2)+1(1)=31; maximum = 40; Authenticity Score = 31/40 x 100 = 77.5.

Score	Interpretation
90-100	Excellent authenticity
75-89	Good, minor concerns
60-74	Moderate citation-authenticity risk
40-59	High citation-authenticity risk
Below 40	Very high citation-authenticity risk

G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

Prediction Accuracy = (Correct genuine/fake predictions / Total predictions) x 100

Correct predictions = 6 Total predictions = 9 Prediction Accuracy = 66.67 %

FINAL LAB 1 RESULTS

Measure	Your Result
Total references in AI paper	9
References deeply audited	9
A: Verified	9
B: Wrong metadata	0
C: Frankenstein	0
D: Fabricated	0
E: Identifier mismatch	0
Authenticity Score	__100__ /100
Prediction Accuracy	__66.67__ %

FINAL INTERPRETATION

1. Did professional-looking citations necessarily turn out to be genuine?

Yes

2. Did all genuine references actually support the claims for which they were cited?

Yes

3. What was the most serious citation-integrity problem you observed?

None

4. What is the single most important lesson you learned from this lab?

Although all citations in my AI generated research paper turned out to be accurate and correct, it might not necessarily be the case for all AI generated papers. Hence it is necessary to cross verify all references provided by AI.

SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_RollNumber_Name

Example: Lab1_IEC2026001_RahulSharma

STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

Student Name: Yuvraj Singh Rajawat

Roll Number: PHM2026001

Signature:

Date: 21/08/2026

CENTRAL LESSON OF LAB 1

Does the source exist? + Does the source actually support the claim?

AI Generation -> Reference Inventory -> Systematic Sampling -> Prediction -> Authenticity Verification -> Claim-Support
Verification -> Quantitative Scores -> Submission

AI-Assisted Citation Integrity Audit | Parts A-G